

The Impact of a Pharmacist-Led Patient Management Program in Inflammatory Disease Patients at a Health-System Owned Specialty Pharmacy (HSSP)

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BACKGROUND

- Inflammatory Disease (ID) is characterized by chronic inflammation as in Ankylosing Spondylitis (AS), Dermatology ID conditions, Inflammatory Bowel Disease (IBD), Juvenile Idiopathic Arthritis (JIA), Psoriatic Arthritis (PSA), Psoriasis (PSO), and Rheumatoid Arthritis (RA).
- Clinical pharmacists in the patient management program (PMP) manage ID by conducting comprehensive initial telehealth assessments (IA), clinical follow-up, disease education, and medication counseling.
- Assessing patient reported outcomes and safety monitoring occur during semiannual reassessments (REA), and monthly refill screenings for adherence are completed for patients enrolled in the program.

OBJECTIVES

This study aims to measure the effectiveness of the PMP in the following patient reported outcomes (PRO): pain and flares (all inflammatory conditions), and RAPID3 scores (in RA).

METHODS

A retrospective pre-post intervention chart review was conducted at Vivo Health Pharmacy, a health-system specialty pharmacy (HSSP) in NY state. Patients with an IA documented since 2021 and with their latest REA completed in 2024 were included. Baseline characteristics and paired comparisons for patients with IA and REA were reported for pain, flares, and RAPID3 (for RA) and severity (for RA), measured as Severe >12, Moderate 6.1-12, Low 3.1-6, Near remission ≤3.

INCLUSION	EXCLUSION
Opted into the PMP	Declined or transitioned out
≥ 2 telehealth assessments	No reassessments documented
Diagnosis of ID	< 3 refills with HSSP

RESULTS

Female sex	59.1%
Age (0-17 years)	12.2%
(18-44 years)	34.1%
(45-64 years)	29.6%
(65-79 years)	11.4%
(>80 years)	12.6%
Months between IA and latest REA (average)	12.6 (SD 7.1)

Table 1. Baseline Characteristics (N=1,313)

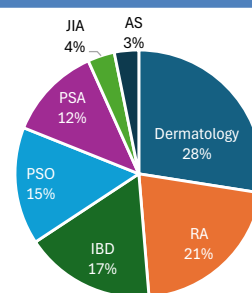


Figure 1. Distribution of all ID (N=1,313)

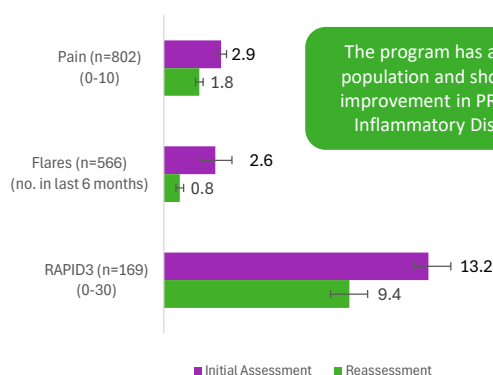


Figure 2. Average Reduction in Flares, Pain, RAPID3 score (all p<0.001)

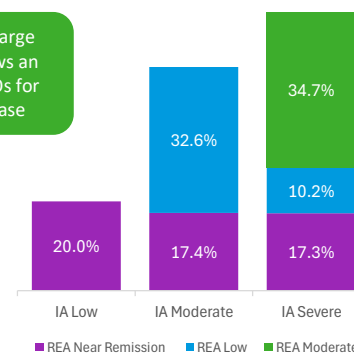


Figure 3. Percent Patients with Improved Severity from IA to REA

DISCUSSION

There were 1,313 patients with ID enrolled in the HSSP PMP included in this analysis. Patient age averaged 51.1 years (SD 30.9), with 59.1% females. PRO from the latest REA were used to compare to their baseline outcomes documented during IA. There was a significant improvement in patient reported pain scores and flares across all disease states. For RA patients, RAPID3 score decreased from 13.2 to 9.4 (p<0.001), with 62.2% of RA patients moving from severe to less-severe RAPID3 score (p<0.001).

Limitations include:

- A focus on short term outcomes, which may not fully capture the effect of the PMP in ID, which are chronic conditions.
- Not all patients are able to quantify or provide PROs at time of assessment, limiting the responses used for paired analyses.
- IAs were completed for new HSSP starts and may have included patients already on therapy but newly enrolled in the PMP.

CONCLUSIONS

The PMP demonstrated significant improvements in PROs for pain and flares across ID conditions and RAPID3 in RA. These results provide strong evidence that integrating specialized clinical pharmacists may result in positive clinical outcomes in the management of complex medication regimens in ID.

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